Final Project Report

Where Did My Money Go?

A Visual Story of U.S. Inflation & Consumer Prices (2015–2025)

Project Scope

For years, every dollar earned has quietly bought less. This project maps out the story of inflation and its real impact on American households between 2015 and 2025.

Using data from the U.S. Bureau of Labor Statistics and the Bureau of Economic Analysis, the project explores three core questions:

- How have consumer prices changed across major spending categories?
- How has inflation evolved year by year?
- How does where you live influence your cost of living?

The United States has one of the largest and most comprehensive economies in the world, making its inflation data particularly rich for analysis. The final deliverables include an interactive-style visual presentation that transforms dry economic data into clear, compelling stories.

Why This Project Is Interesting to Me

The value of money is always changing, and it is both fascinating and a little shocking to see how much prices have soared over the past decade. From rent and groceries to fuel and everyday essentials, the increases have been very real.

This project turns the abstract concept of "inflation" into something tangible and visual. By creating clear side-by-side comparisons from 2015 to 2025, it makes the changes very visible.

The project also highlights how individuals experience inflation differently in their daily lives. Depending on what you buy and where you live, your personal "inflation rate" can feel dramatically higher or lower than the official national numbers. A renter in a big city might feel the squeeze far more than a homeowner with a fixed mortgage, while a family with heavy grocery bills experiences it differently from someone who drives frequently.

Data Sources & Methodology

Consumer Price Index (CPI) Data

- **Source**: U.S. Bureau of Labor Statistics (BLS)

- The CPI measures the average change over time in the prices paid by urban consumers for a market basket of consumer goods and services.
- Originally downloaded as separate Excel files by category and year, I cleaned and combined them into one unified dataset covering 2015 through 2025.
- Major categories analyzed: Housing (Shelter), Transportation (including fuel), Food, Medical Care, and Recreation.

Regional Price Parities (RPP) Data

- **Source**: Bureau of Economic Analysis (BEA)
- This dataset provides Regional Price Parity values for every U.S. state from 2014 to 2024.
- RPP shows how expensive it is to live in each state relative to the national average (set at 100).
    - RPP > 100 = more expensive than average
    - RPP < 100 = cheaper than average

Key Visualizations & Findings

1. How Much Is $100 from 2015 Worth Today?

This visualization shows the real purchasing power decline. Due to cumulative inflation, $100 in 2015 buys significantly less in 2025 across most categories.

2. Which Spending Areas Were Hit the Hardest?

- **Housing** was impacted the most, rising dramatically over the decade.
- **Food** and **Transportation (fuel)** showed sharp increases, especially during the 2021–2022 surge.
- **Recreation** experienced the least inflation pressure.

The timeline visualization uses creative icons that "drain" as inflation reduces what $1 can buy. In particular, **$1 buys only about 68 cents worth of housing** today compared to 2015.

3. Where You Live Matters

Using Regional Price Parities, the project demonstrates significant geographic variation in cost of living. Coastal states and major metropolitan areas consistently show RPP values well above 100, while many Midwest and Southern states remain below the national average.

4. Timeline of Inflation (2015–2025)

Inflation was mostly dormant pre-2020 but proved highly sensitive to major economic shocks. A dramatic spike occurred in 2021–2022, driven by global supply chain disruptions after

COVID-19, massive fiscal stimulus, and the energy shock from the war in Ukraine. Prices have since moderated but remain above the Federal Reserve's 2% target.

Key Insight: Energy, food, and housing (shelter) remain the biggest drivers of overall inflation.

Design & Presentation Approach

The presentation was intentionally designed to be visually engaging and suitable for a professional portfolio. It uses:

- Clean layout and consistent color scheme
- Before-and-after style comparisons
- Creative icon-based visualizations (draining dollar icons)
- Clear charts contrasting category-specific inflation with overall CPI
- Geographic maps showing cost-of-living differences

The two key charts you included contrast each other effectively: one highlights the dramatic price surges in rent, groceries, and fuel, while the other provides broader context on overall inflation trends.

Conclusion

This project illustrates that inflation is not just an abstract economic statistic,it has tangible, uneven effects on American households. Housing has been the hardest-hit category over the past decade, followed by food and energy. While overall inflation has cooled since the 2022 peak, the cumulative impact from 2015 to 2025 has meaningfully reduced purchasing power for many families.

By combining national CPI trends with regional price data and creative visualizations, this project makes complex economic realities accessible and memorable. Understanding these patterns is increasingly important in today's economy.